

# Canberk Baykal

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## EDUCATION

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| <b>University of Cambridge</b> | Oct 2023 - <i>now</i> , Cambridge, UK<br>Ph.D. in Computer Science<br><b>Research Areas:</b> Computer Vision, Computer Graphics                                                                                                                                                                                                                              |
| <b>Koç University</b>          | Sep 2020 - June 2022, İstanbul, Türkiye<br>M.Sc. in Computer Science and Engineering<br><b>CGPA:</b> 4.00 / 4.00<br><b>Research Areas:</b> Computer Vision, Natural Language Processing<br><b>Relevant Coursework:</b> Deep Learning, Computer Vision with Deep Learning, Natural Language Processing, Deep Unsupervised Learning, Advances in Deep Learning |
| <b>Bilkent University</b>      | Sep 2016 - June 2020, Ankara, Türkiye<br>B.S. in Electrical and Electronics<br><b>CGPA:</b> 3.70 / 4.00<br><b>Relevant Coursework:</b> Introduction to Machine Learning, Artificial Intelligence, Neural Networks                                                                                                                                            |

## RESEARCH EXPERIENCE

My current research interests lie in 3D Computer Vision, including 3D capture and reconstruction. I am also interested in image generation and editing as well as natural language processing.

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|----------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------|
| <b>Adobe Research</b><br>(London, UK)                          | 06/2025 – 01/2026    | <b>Research Scientist Intern</b><br>Supervisor: Iliyan Georgiev                       |
| <b>University of Cambridge</b><br>CORE Lab<br>(Cambridge, UK)  | 10/2023 – <i>now</i> | <b>PhD Researcher</b><br>Supervisor: Prof. Cengiz Öztireli                            |
| <b>Koç University</b><br>KUIS AI Center<br>(İstanbul, Türkiye) | 09/2020 – 09/2023    | <b>Graduate Researcher</b><br>Supervisors: Prof. Aykut Erdem<br>and Prof. Deniz Yuret |

## TEACHING EXPERIENCE

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| <b>University of Cambridge</b><br>Computer Science Department<br>(Cambridge, UK) | 10/2023 – now     | <b>Supervisor</b><br>Machine Learning and Bayesian Inference                                                                                                                                                             |
| <b>Koç University</b><br>KUIS AI Center<br>(İstanbul, Türkiye)                   | 09/2020 – 09/2023 | <b>Teaching Assistant</b><br>COMP106 - Discrete Mathematics<br>for Computer Science and Engineering<br>COMP547 - Deep Unsupervised Learning<br>COMP202 - Data Structures & Algorithms<br>COMP527 - Computational Imaging |

## INDUSTRY EXPERIENCE

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|-------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Aselsan</b><br>(Ankara, Türkiye)             | 07/2019 – 08/2019 | <b>Summer Intern</b><br>VHDL Design and Verification Techniques                                                                                               |
| <b>INFINIA Mühendislik</b><br>(Ankara, Türkiye) | 06/2018 – 08/2018 | <b>Summer Intern</b><br>LED controls on a Model Automation<br>System using Raspberry Pi 3 Model B+<br>and LED control interface using<br>ESP8266 Wi-Fi Module |

## PUBLICATIONS AND PAPERS

- ResEdit: Residual embeddings for precise generative image editing. **Ahmet Canberk Baykal**, Valentin Deschaintre, Yannick Hold-Geoffroy, Michael Fischer, Anna Frühstück, Cengiz Öztireli, Iliyan Georgiev. In *Eurographics Symposium on Rendering (EGSR)*, 2026.
- Quartet of Diffusions: Structure-Aware Point Cloud Generation through Part and Symmetry Guidance. Chenliang Zhou, Fangcheng Zhong, Weihao Xia, Albert Miao, **Canberk Baykal**, Cengiz Oztireli. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2026.
- HyperGAN-CLIP: A Unified Framework for Domain Adaptation, Image Synthesis and Manipulation. Abdul Basit Anees, **Ahmet Canberk Baykal**, Muhammed Burak Kizil, Duygu Ceylan, Erkut Erdem, Aykut Erdem. In *SIGGRAPH Asia 2024 Conference Papers*, 2024.
- CLIP-guided StyleGAN Inversion for Text-driven Real Image Editing. **Ahmet Canberk Baykal**, Abdul Basit Anees, Duygu Ceylan, Erkut Erdem, Aykut Erdem, Deniz Yuret. In *ACM Transactions on Graphics*, 2023. Presented in *SIGGRAPH Asia 2023*.

## SKILLS

|                              |                                                                       |
|------------------------------|-----------------------------------------------------------------------|
| <b>Programming Languages</b> | Python, Java, C++, Julia                                              |
| <b>Frameworks</b>            | PyTorch, Keras, Tensorflow, Knet (Julia)                              |
| <b>Developer Tools</b>       | Git, PyCharm, Spyder, Jupyter, IntelliJ, VSCode, MATLAB               |
| <b>Languages</b>             | Turkish (C2), English (TOEFL iBT: 110, C2), German (A2), Spanish (A1) |

## HONORS AND AWARDS

- Cambridge Trust & Computer Science Premium Scholarship, 2023-2026.
- TensorFlow Developer Certificate, 2022-2025.
- AI Fellowship, KUIS AI Center, 2020-2023.
- TÜBİTAK National M.Sc. Scholarship Program, 2020-2022.
- 2<sup>nd</sup> place in IEEEExtreme Turkey Camp Contest, 2019.
- 19th place in Turkish Programming Contest (regional elimination round of ICPC), 2019.
- ALES (Selection Exam for Academic Personnel and Graduate Studies), Score: 96, Rank: 230 out of 122.927, 2019.
- Bilkent University Comprehensive Scholarship. Covers 100% of tuition and monthly stipends during the study, 2016-2020.
- Scholarship of the Turkish Prime Ministry. Monthly stipends for outstanding performance in the Nationwide University Entrance Exam LYS, 2016-2020.
- Ranked 38th in the Nationwide University Entrance Exam LYS and 51st in YGS among 2 million students in Turkey, 2016.

## ACADEMIC SERVICE

- Reviewer, ICIP 2024, 2025.
- Reviewer, SIGGRAPH 2023.

## EXTRACURRICULAR ACTIVITIES

- Lucy Cavendish College Boat Club, University of Cambridge, Rower, 2023-2024.
- Bilkent Cycling Society, General Secretary, 2018-2019.
- Bilkent IEEE Student Branch, Active Member, 2017.